

MATTHEW WONG

(925) 786-3774 • Math.You.TW@gmail.com • [Github](#) • [Portfolio](#) • [LinkedIn](#)

SUMMARY

Aspiring data-driven Data Science graduate of UC San Diego with a strong foundation in statistical modeling and natural language processing. Experienced in Python, SQL and R to clean, analyze, and visualize structured and unstructured data. Developed a containerized GenAI pipeline that tripled data extraction yield. Proven leader adept at translating complex datasets into actionable insights.

EDUCATION

University of California, San Diego | *B.S. Data Science (GPA: 3.79)* 2023 - 2026

- **Coursework:** Probabilistic Modeling & ML, Representation Learning, Systems for Scalable Analytics, Statistical NLP, Data Analysis & Inference, Data Science in Practice

Diablo Valley College | *Pleasant Hill, CA* | *Computer Science Coursework* 2021 - 2023

EXPERIENCE

Sharetea | *Assistant Manager* | *San Ramon, CA* 2021-2023

- Organized weekly inventory logistics by coordinating with 3+ vendors to ensure consistent stock levels.
- Facilitated recruitment by conducting interviews for prospective employees.
- Managed quality control for a 10+ person team, achieving full compliance on internal store inspections.
- Balanced customer service and supervisory duties during 6+ hour peak shifts, demonstrating strong multitasking and team leadership.

East Bay Field Hockey | *Coach and Team Member*, *San Ramon, CA* 2015-2021

- Resolved logistical and training issues for 15+ athletes per season, maintaining high practice attendance and cohesion.
- Worked closely with the Head Coach through regular check-ins to monitor development plans and retention goals, resulting in improved athlete growth and increased retention rates.

PROJECTS

Market Foundry | [Market Foundry](#) March 2026

- Built a schema-guided, model-agnostic knowledge extraction pipeline for financial text (SEC filings, earnings calls, news, research) that generates structured causal triples in Neo4j.
- Designed and implemented a constraint-based YAML schema system to enforce valid entities and relationships, reducing unsupported or spurious causal links.
- Containerized the pipeline with Docker and integrated it with Neo4j to support scalable ingestion and querying of large document collections.
- Enabled structured querying of causal claims across documents (e.g., how specific events, risks, or decisions propagate through companies or sectors), supporting downstream analytical and research workflows.
- **Tools:** Python, Neo4j, Docker, Git, YAML

League of Legends Objective Analysis | [League of Legends Objective Analysis](#) Dec 2024

- Conducted a statistical study of professional match data to quantify how early objective control influences win probability.
- Assessed data quality via hypothesis tests on missingness to determine whether imputation was necessary before modeling.
- Trained a baseline decision tree to predict match outcomes, then improved performance using a tuned random forest via cross-validation.
- Quantified the win-rate impact of securing early objectives to provide evidence-based recommendations for strategic decision-making in competitive play.
- **Tools:** Python, Pandas, NumPy, SciPy, Statsmodels, Scikit-learn, Matplotlib, Seaborn

SKILLS

- **Technical Skills:** Data Analysis, Machine Learning (Scikit-learn), Statistical Modeling, Natural Language Processing (NLP), Data Visualization (Matplotlib, Seaborn), Knowledge Graph Engineering, Hypothesis Testing, Cross-Validation, Data Collection, Data Cleaning, Quantitative Analysis, Qualitative Analysis
- **Tools & Technologies:** Python, Pandas, R, SQL, Neo4j, Docker, Git, Tableau, Excel, YAML
- **Soft Skills:** Leadership, Data Storytelling, Communication, Critical Thinking, Adaptability, Self-Directed Learning